

DSA210 Term Project — Final Report

*Identifying High-Potential but Under-Engaged Customers
Through Education, Income, Campaign Behavior, and Macroeconomic
Context*

DSA210 — Introduction to Data Science
2025–2026 Spring Term
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1. Motivation

Marketing campaigns are expensive, and their effectiveness varies widely across customer segments. A common assumption in retail analytics is that higher education correlates with greater spending power, and therefore, with higher campaign engagement. This project challenges that assumption.

The central observation motivating this work is: although higher education is associated with greater absolute spending, campaign engagement does not scale uniformly across education tiers. In fact, the highest concentration of high-spending but campaign-unresponsive customers — the so-called 'TargetSegment' — appears among lower education groups and among high-earning customers who simply ignore marketing outreach.

Research Question: *Although higher education is associated with greater spending, campaign engagement does not scale uniformly across education tiers — and the highest concentration of high-spending but non-responsive customers is found among lower education groups. This project investigates whether education level, relative income, and macroeconomic context at enrollment (CPI) can jointly explain campaign disengagement and identify the most untapped customer segments.*

2. Data Source

2.1 Marketing Campaign Dataset

The primary dataset is the Customer Personality Analysis dataset from Kaggle (originally sourced from Dr. Omar Romero-Hernandez's work). It contains 2,240 customers with 29 features covering demographics, product spending, and campaign response history.

Attribute	Description	Example Values
Year_Birth, Education, Income	Demographics	PhD, \$60,000
MntWines ... MntGoldProds (6 cols)	Product spending (2 yrs)	\$0–\$1,500
AcceptedCmp1–5, Response	Campaign responses (binary)	0 / 1
NumWebPurchases, NumStorePurchases, ...	Purchase channels	0–28 purchases
Kidhome, Teenhome	Children at home	0–2

2.2 US CPI (Inflation) Dataset

The secondary dataset contains monthly US CPI values from FRED/Kaggle, covering 1947–2023 (918 observations). Each customer was matched to the CPI value of their enrollment month (Dt_Customer), providing macroeconomic context at the time of acquisition.

3. Data Analysis

3.1 Data Preprocessing

- Loaded marketing_campaign.csv (tab-separated, 2,240 rows) and US_inflation_rates.csv (918 rows).
- Parsed dates and created a month-level key for CPI merging — resulting in zero null CPI values after the merge.
- Imputed 24 missing Income values using within-education-group median imputation.
- Removed 3 rows with Age > 100 (clearly erroneous birth years).
- Final cleaned dataset: 2,237 customers, 43 features.

3.2 Feature Engineering

The following engineered features were central to the analysis:

Feature	Formula / Logic	Purpose
TotalSpend	Sum of 6 product spend columns	Overall spending power
CampaignScore	Sum of 6 campaign acceptance cols (0–6)	Engagement level
CampaignEngaged	Binary: CampaignScore > 0	Engaged or not
Frequency	Total purchases across all channels	Purchase activity (RFM)
RelativeIncome	Income / median income of same education group	Within-peer income standing
EduTier	Ordinal encoding of education (0–4)	ML-ready education feature
HighPotential	TotalSpend ≥ 75th pct within education group	High spender flag
TargetSegment	HighPotential = 1 AND CampaignScore = 0	Primary target: high-spend, non-responsive

3.3 Exploratory Data Analysis

Key EDA findings:

- Education distribution: Graduation dominates (50.1%), followed by PhD (20.2%), Master (17.0%), 2n Cycle (10.6%), Basic (2.1%).
- Spending increases with education: PhD customers have the highest median total spend (~\$570 vs. ~\$60 for Basic).
- Campaign engagement also increases with education: PhD engagement rate $\approx 31\%$, Basic $\approx 7\%$. However, engagement never reaches majority, leaving most customers unresponsive regardless of education.
- TargetSegment concentration: Basic has the highest TargetSegment rate (24.1%), versus PhD (10.9%), suggesting lower-education high-spenders are particularly untapped.
- CPI at enrollment is slightly lower for engaged customers (median ≈ 232.9) vs. non-engaged (median ≈ 233.2). While the difference is statistically significant (H5), the effect size is negligible — this association should not be interpreted causally; unmeasured confounders (e.g. time-period effects, cohort characteristics) likely explain much of the gap.
- Correlation heatmap: TotalSpend correlates strongly with Income (0.64) and Frequency (0.72). HasChild shows a strong negative correlation with TotalSpend (-0.52), indicating child-free customers spend significantly more.

3.4 Hypothesis Testing

Ten statistical hypothesis tests were conducted to systematically validate the research question. All tests used a significance level of $\alpha = 0.05$.

#	Hypothesis	Test	p-value	Result
H1	Education level affects campaign engagement	Chi-square	0.0064	✓ Significant
H2	PhD customers spend more than non-PhD	Mann-Whitney U	0.0028	✓ Significant
H3	Among PhD: non-engaged spend more than engaged	Mann-Whitney U	1.0000	✗ Not significant
H4	Among PhD: non-engaged have higher relative income	Mann-Whitney U	1.0000	✗ Not significant
H5	CPI at enrollment differs by engagement	Mann-Whitney U	0.0001	✓ Significant
H6	Campaign score differs across education groups	Kruskal-Wallis	0.0032	✓ Significant
H7	TargetSegment rate differs across education groups	Chi-square	0.0774	✗ Not significant
H8	Low-edu high-spenders engage less than high-edu	Mann-Whitney U	0.0154	✓ Significant
H9	Having children is associated with TargetSegment	Chi-square	<0.0001	✓ Significant
H1	Total spend differs across all education	Kruskal-	<0.0001	✓ Significant

#	Hypothesis	Test	p-value	Result
0	groups	Wallis		

7 out of 10 hypotheses were confirmed. The non-significant results for H3 and H4 are themselves informative: within PhD customers, campaign engagement is not explained by income level — engagement is associated with other factors (preferences, channels, lifestyle), not financial capacity alone.

4. Machine Learning Methods

4.1 Problem Formulation

The ML task was framed as a binary classification problem: predict whether a customer belongs to the TargetSegment (high-potential & under-engaged). The dataset contains 2,237 records with a class imbalance of approximately 11.8% positive (TargetSegment) vs. 88.2% negative.

Features used: EduTier, Income, RelativeIncome, Age, TotalSpend, Frequency, Recency, CPI_at_enrollment, HasChild (9 features). The train/test split was 80/20 with stratification to preserve class balance.

4.2 Classification Models

Four classifiers were trained and evaluated using ROC-AUC as the primary metric (appropriate for imbalanced classification). All models used `class_weight='balanced'` or equivalent to handle the 12% minority class.

Model	Test ROC-AUC	CV ROC-AUC (5-fold)	Notes
Logistic Regression (baseline)	0.8711	0.8853 $\pm$ 0.013	Scaled features
Decision Tree (max_depth=5)	0.8587	0.8997 $\pm$ 0.030	Interpretable
Random Forest (200 trees)	0.9410	0.9463 $\pm$ 0.008	Best model
Gradient Boosting (200 est.)	0.9425	0.9453 $\pm$ 0.010	Close second

Random Forest was selected as the best model based on its combination of high test AUC (0.941), stable cross-validation performance (low variance $\pm$ 0.008), and interpretable feature importances.

4.3 Feature Importances

Random Forest feature importances (Gini-based):

- TotalSpend: 41.7% — the dominant predictor. High-spending customers who do not engage are definitionally the TargetSegment.
- RelativeIncome: 21.9% — customers who earn above their education peer median are more likely to be under-engaged.
- Income: 14.9% — absolute income further distinguishes the segment.
- Frequency: 7.4% — purchase frequency contributes; high-frequency, low-engagement customers form a distinct pattern.
- EduTier: 4.1% — education level has a moderate effect after income and spend are controlled for.
- Recency, Age, CPI_at_enrollment, HasChild: together contribute ~8% of importance.

4.4 K-Means Clustering (Unsupervised)

K-Means clustering (k=4, selected via elbow method) was applied to validate the TargetSegment definition through unsupervised discovery. Clusters were profiled on 8 features.

Cluster	TotalSpend	CampaignScore	Income	RelativeIncome	Target %
0	\$1,584	2.87	\$83,144	1.59	0.0%
1	\$114	0.19	\$34,803	0.70	1.1%
2	\$832	0.28	\$62,183	1.19	16.9%
3 ★	\$1,249	0.37	\$73,719	1.42	40.7%

Cluster 3 emerges as the primary under-engaged high-spender group (40.7% TargetSegment rate). It is characterised by high spend (~\$1,249), near-zero campaign engagement (score 0.37), above-median income (\$73,719), above-peer relative income (1.42), and no children — all consistent with the project's theoretical framework.

PCA visualisation confirms that TargetSegment members cluster distinctly in the high-spend, low-engagement corner of the feature space.

5. Findings

5.1 Key Findings

- Education and spending: PhD customers spend $\sim 9.5\times$ more than Basic customers (median), confirming the education-spending link (H2, H10 both significant).
- Education and engagement: Campaign engagement rates scale with education (Basic 7% $\rightarrow$ PhD 31%), but majority of even PhD customers never respond to campaigns (H1, H6).
- The 'engagement gap': Among high-spending customers, lower-education groups engage significantly less with campaigns than higher-education groups (H8, $p=0.015$). This confirms the core hypothesis that education shapes engagement independently of spending.
- Macroeconomic context: Customers enrolled during higher-CPI periods are associated with marginally lower engagement (H5, $p=0.0001$). This is statistically significant but practically negligible (median difference: 0.3 CPI points); the association should not be interpreted as a causal relationship.
- Children as disengagement predictor: Having children is strongly associated with being in the TargetSegment (H9, $p<0.0001$). Child-free, high-income customers are the least engaged despite highest spending.
- ML performance: Random Forest achieves ROC-AUC = 0.941 (CV: 0.946 ± 0.008), indicating the TargetSegment is highly predictable from available features. Spend and relative income are the dominant predictors.
- Clustering validation: K-Means independently identifies a cluster (Cluster 3) with 40.7% TargetSegment rate, confirming the supervised classification findings through unsupervised means.

5.2 Business Implication

The 263 TargetSegment customers (11.8% of the base) represent the highest-value marketing opportunity: they already spend significantly but have never responded to a campaign. A targeted re-engagement strategy for this segment — particularly Cluster 3 (child-free, high-income, high-spend) — could yield substantial ROI compared to broad-based campaigns.

6. Limitations and Future Work

6.1 Limitations

- Geographic scope: The US CPI enrichment assumes all customers are US-based, which may not hold. The dataset provides no country information, so this is an approximation.
- Temporal static snapshot: The marketing dataset represents a fixed window. Customer behavior likely evolves over time; a longitudinal dataset would be more appropriate for causal inference.
- Class imbalance: TargetSegment at 11.8% required careful handling. Despite balanced class weights, minority-class precision remains $\sim 53\%$ for Random Forest — meaning roughly half of predicted TargetSegment customers are false positives. For a marketing deployment this matters more than AUC: acting on every positive prediction would waste

~half the campaign budget on non-targets. Future work should optimise the classification threshold to balance precision and recall based on campaign cost-per-contact vs. expected revenue per converted customer.

- Causal inference not possible: Statistical associations are established, but no causal claims can be made from this observational data. Higher education does not necessarily cause lower campaign engagement.
- H3 and H4 non-significant: The hypotheses that non-engaged PhD customers spend more or have higher relative income were not supported. This may indicate that within-education-group spend variation does not drive the PhD engagement gap — external factors (time constraints, channel preferences) may be more relevant.

6.2 Future Work

- Channel analysis: Investigate whether specific campaign channels (digital vs. print vs. in-store) drive the engagement gap, especially for lower-education high-spenders.
- Time-series modelling: If transaction-level data were available, RFM trends over time could yield more actionable segmentation.
- A/B testing: Validate the TargetSegment definition against actual campaign outcomes to confirm that these customers are genuinely reachable with the right message.
- Richer feature engineering: Incorporate product category preferences to personalise outreach within the TargetSegment.
- Neural network / XGBoost comparison: More expressive models may improve minority-class precision, which remains a limiting factor.

7. AI Tool Disclosure

This project used Claude (Anthropic) as an AI assistant in the following capacities, consistent with the course guidelines:

- Code structure suggestions: Initial scaffolding of the preprocessing, EDA, hypothesis testing, and ML pipeline scripts.
- Statistical test selection: Guidance on choosing non-parametric tests (Mann-Whitney U, Kruskal-Wallis) given non-normal distributions.
- Documentation: Docstrings and README formatting.
- Report writing: Final report drafted with Claude assistance; all analytical decisions, numerical results, and interpretations were reviewed and validated by the student.

All code was executed locally by the student and outputs verified. No LLM was used to generate or fabricate results.

GitHub Repository: <https://github.com/emiralpaktar/dsa210>